INCH-POUND

MIL-C-24643/48D
22 November 1994
SUPERSEDING
MIL-C-24643/48C
14 March 1994

MILITARY SPECIFICATION SHEET

* CABLE, ELECTRICAL, 1000 VOLTS, TWO CONDUCTOR TYPE LSDNW
(INCLUDING VARIATION LSDNWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LSDNW (unarmored), and LSDNWA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, uncoated (see table I for size).
Second - Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation or ethylene propylene rubber insulation (see table I for thickness). Standard identification code shall be applied by method 1 or method 3. If identification method 1 is used, insulation color shall be black.
Fourth - Two conductors cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well rounded assembly.
* Fifth - Binder tape applied helically with overlap.

Construction variant for type LSDNW

Sixth - Cross-linked polyolefin jacket (see table I for thickness).


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Construction variant for type LSDNWA

Sixth - Cross-linked polyolefin jacket (see table I for thickness).
Seventh - Braided metal armor

TABLE I. Details.

| Military 1/ part no. M24643/48 | Size 2/ | AWG conductor size | Insulation thickness min avg (inch) | Cable jacket thickness min avg (inch) | Overall diameter (inches) | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | TYPE LSDNW | | TYPE LSDNWA | |
| | | | | | min | max | min | max |
| -01UN | 3 | 16 (class B) | 0.030 | 0.040 | 0.361 | 0.390 | 0.411 | 0.440 |
| -02UN | 4 | 14 (class B) | .030 | .040 | .398 | .430 | .448 | .480 |
| -03UN | 9 | 10 (class B) | .030 | .040 | .504 | .545 | .554 | .595 |
| -04UN | 14 | 9 (class B) | .045 | .040 | .564 | .610 | .614 | .660 |
| -05UN | 23 | 7 (class B) | .045 | .045 | .638 | .690 | .688 | .740 |
| -06UN | 50 | 3 (class C) | .045 | .050 | .842 | .910 | .892 | .960 |
| -07UN | 75 | 1 (class C) | .055 | .055 | 1.000 | 1.080 | 1.050 | 1.130 |
| -08UN | 100 | 0 (class D) | .055 | .055 | 1.082 | 1.170 | 1.132 | 1.220 |

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LSDNW and LSDNWA.

TABLE II. Electrical details.

| Size 1/ | AWG conductor size | Conductor resistance (ohms) | Insulation resistance (megohms) | Voltage Withstand (minimum) (volts) | Cold Bending mandrel diameter (max inch) | Accelerated service loading current (amperes) |
|---|---|---|---|---|---|---|
| 3 | 16 (class B) | 4.3 | 6000 | 4000 | 4 | ---- |
| 4 | 14 (class B) | 2.68 | 5000 | 4000 | 4 | ---- |
| 9 | 10 (class B) | 1.08 | 3800 | 4000 | 5 | ---- |
| 14 | 9 (class B) | .86 | 4400 | 5500 | 6 | 100 |
| 23 | 7 (class B) | .54 | 3700 | 5500 | 7 | 130 |
| 50 | 3 (class C) | .21 | 2600 | 5500 | 9 | 215 |
| 75 | 1 (class C) | .13 | 2600 | 7000 | 10 | 285 |
| 100 | 0 (class D) | .11 | 2400 | 7000 | 11 | 335 |

1/ Type LSDNW and LSDNWA.

EXAMINATION AND TESTS:

Requirements

Basic electrical:

Conductor Resistance - ohms/1000 feet at 25°C, max ..... (see table II)
Voltage Withstand - volts, root mean square, minimum ..... (see table II)
Insulation resistance - Conductor to conductor
Megohms/1000 feet minimum ..... (see table II)
Conductor continuity ..... No failure



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EXAMINATION AND TESTS (continued):

| | Requirements |
|---|---|
| Group A: | |
| Visual and dimensional | No failure |
| Group B: | |
| Cross-linked proof test (percent, maximum) | |
| Insulation | 50 |
| Jacket (When tested at 200°C) | 50 |
| Cold bending, cable - (see table I for mandrel diameter) | No damage |
| Tear - pounds per inch thickness, minimum (ASTM D 470) | 35 |
| Physicals (unaged) | |
| Insulation | |
| Ethylene propylene rubber | |
| Tensile strength - $lb/in^2$, minimum | 700 |
| Elongation - percent, minimum | 250 |
| Cross-linked polyethylene | |
| Tensile strength - $lb/in^2$, minimum | 1800 |
| Elongation - percent, minimum | 250 |
| Jacket (cable) | |
| Tensile strength - $lb/in^2$, minimum | 1300 |
| Elongation - percent, minimum | 160 |
| Group C: | |
| Physicals (aged) | |
| Insulation | |
| Ethylene propylene | |
| Air Oven | |
| Tensile strength - percent of unaged, minimum | 75 |
| Elongation - percent of unaged minimum | 75 |
| Cross-linked polyethylene | |
| Air Oven | |
| Tensile strength - percent of unaged, minimum | 80 |
| Elongation - percent of unaged minimum | 80 |
| Jacket (cable) | |
| Air oven | |
| Tensile strength - percent of unaged, minimum | 60 |
| Elongation - percent of unaged minimum | 60 |
| Hot oil immersion | |
| Tensile strength - percent of unaged, minimum | 50 |
| Elongation - percent of unaged minimum | 50 |



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EXAMINATION AND TESTS (continued):

| | Requirements |
|---|---|
| Shrinkage | No failure |
| Heat distortion - percent of unaged, maximum | 30 |
| Permanence of printing (conductor) - method 1 only, cycles, minimum | 25 |
| * Permanence of printing (jacket) - cycles, minimum | 125 |
| Armor type LSDNWA only - conformance to material, construction and coverage | No failure |

Group D:

| | |
|---|---|
| Flame propagation (cable) | No failure |

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

| | Requirements |
|---|---|
| Aging and compatibility (cable) (125 ± 5°C) | No failure |
| Abrasion resistance (jacket) - scrapes, minimum | 75 |
| Acid gas equivalent - percent, maximum | |
| Jacket | 2 |
| Fillers | 2 |
| Insulation | 18 |
| Halogen content - percent, maximum | |
| Jacket | 0.2 |
| Fillers | 0.2 |
| Immersion (jacket) | |
| Tensile strength - percent of unaged, minimum | 50 |
| Elongation - percent of unaged, minimum | 50 |
| Smoke index, maximum | |
| Jacket | 25 |
| Fillers | 45 |
| Insulation | 45 |
| Toxicity index, maximum | |
| Jacket | 5 |
| Fillers | 5 |
| Insulation | 1.5 |
| Durometer (jacket) - (type A) hardness, minimum | 80 |
| Weathering (jacket) | No failure |

UNIT ORDERING LENGTHS:

| Type and Size | Feet (nominal) |
|---|---|
| LSDNW-3 | 2500 |
| LSDNW-4 through 23 | 2000 |
| LSDNW-50 | 1500 |
| LSDNW-75, 100 | 1000 |

Note: Not for AIR FORCE use.

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